



Sprite III (Alpha) Quick Start Guide

Overview

This guide is intended for those who are familiar with installing and using dispensing systems. It provides a quick overview of the installation and use of the Sprite III (Alpha), with brief instructions for each step. For more complete information, please refer to the Sprite III (Alpha) Reference Manual (p/n 20-09070-00). To obtain a copy, please call customer service at 1-800-543-7184 or download it at www.hydrosystemsco.com Brands > Nova.

Follow these steps to begin using the Sprite III Alpha:

1. Install mechanical and electrical portions of the system. Refer to “*Mechanical Installation*” on page 2 and “*Electrical Installation*” on page 2 for instructions.
2. Familiarize yourself with the three keys on the keypad and with the User menus.
3. Within the Program Mode menus, familiarize yourself with the Configuration Settings and Adjustment Menu (Probe or Probeless) that you will use. Note the acronym used for each menu is in *Figure QS-8 on page 5*.
4. Input the Installer Password.
5. Configure the system’s settings (see key icons in *Figure QS-5 ‘Controller Keys’ on page 2*). Press the **SCROLL** key to change the display to **CONF** (Configuration Settings). Your configuration settings determine how your system operates and which menus are available to you during programming. Choose between:
 - Detergent control (Probe or Probeless)
 - Machine type (Door or Conveyor)
 - Detergent type (Liquid or Capsule)
 - Third-product pump selection (with Rinse, with Detergent, or disabled).

After setting the configuration, you need only program the adjustments that apply to your system.

6. Press **SCROLL** to change the display to **ADJ.P** (Probe Adjustments) or **ADJ.n** (Probeless Adjustments). Set adjustments as needed.
7. Press **ENTER** to return to the main menu. To exit, wait 30 seconds; the menu will automatically return to the Home screen after 30 seconds of inactivity.
8. Refer to the Sprite III (Alpha) Reference Manual (p/n 20-09070-00) for more detailed information.

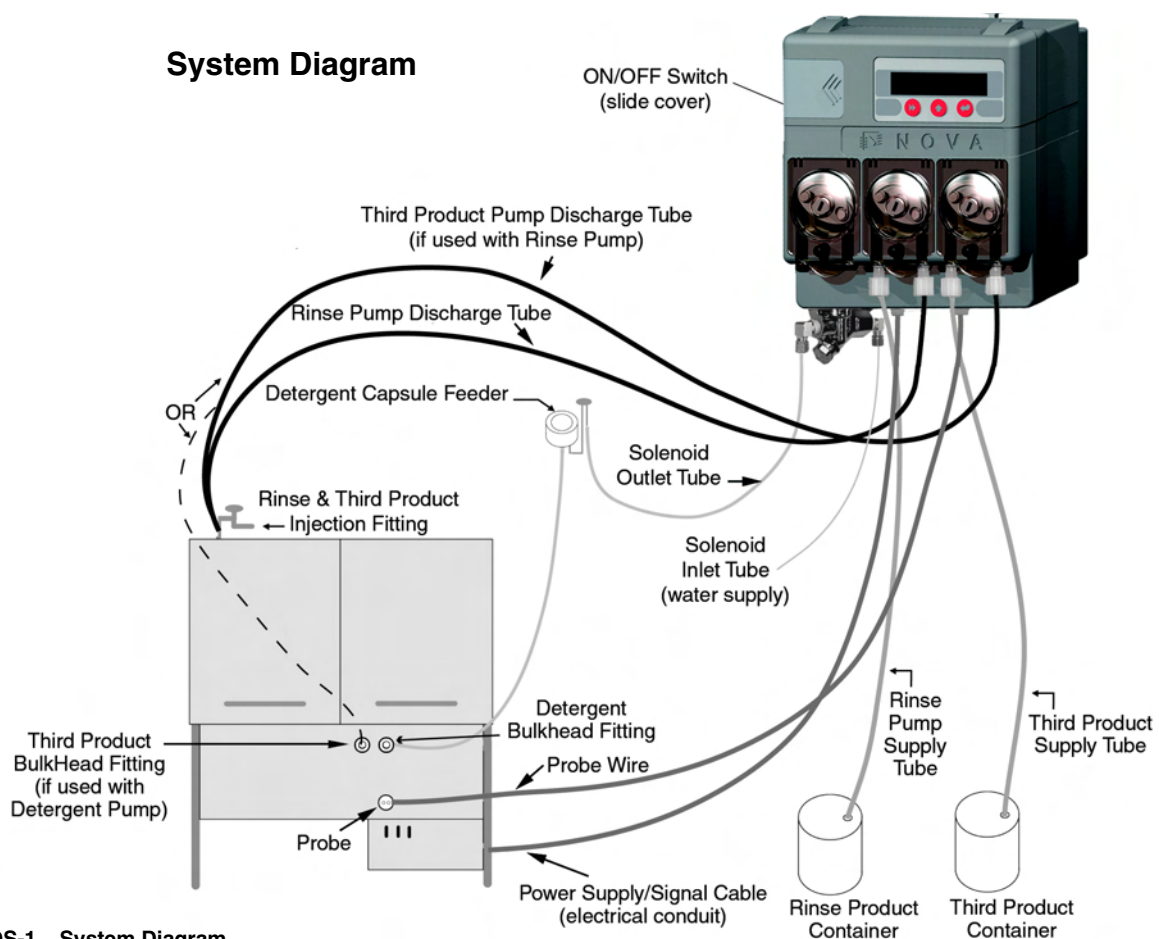


Figure QS-1 System Diagram

Mechanical Installation

Caution



Disconnect power to dish machine and dispenser during installation or any time cabinet is opened.

Wall Mount Dispenser

1. Locate dispenser near product containers, at a height for easy service access.
2. Place, level and install mounting bracket.
3. Lower dispenser onto bracket. Press down to lock.
4. Install new probe (for probe operation only). Immerse the probe in the wash tank solution near the product entry point and ensure a good solution flow.
5. Install the Rinse Injection Fitting at least 6" below dish machine's rinse plumbing vacuum breaker. To prevent leaks, use thread sealant.

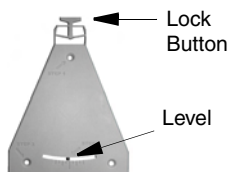


Figure QS-2
Mounting Bracket

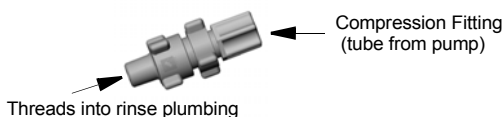


Figure QS-3 Rinse Injection Fitting

6. Mount the Detergent Bulkhead Fitting above water line in tank and as close to probe as possible.
7. Route detergent, rinse and third-product supply tubes as indicated in 'System Diagram' on page 1.

Caution



Use only copper tubing if water source is hot. **DO NOT** connect solenoid to the boosted temperature rinse water line on high temperature dish machines.

Electrical Installation

Caution



Disconnect power to the machine and dispenser during installation or whenever cabinet is opened. Verify all voltage sources with a meter. Refer to Figure QS-4, "Wiring Diagrams."

All electrical connections (except the probe) must be made either in the dish machine control circuit panel or an external junction box. The dispenser is pre-wired with a multi-conductor electrical cable that must be run through a conduit to the hard-wired connection on the dish machine.

Use a 1/2" (13 mm) ID flexible, water tight conduit that meets all local and national codes. A conduit fitting is on the bottom of the dispenser, where the power cable exits. The probe wire is also pre-wired and must be routed to the probe location and cut to fit.

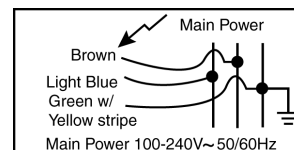
Caution



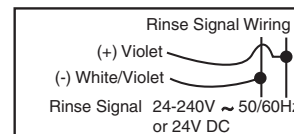
For **3-Phase Power System Connections**, refer to the Sprite III (Alpha) Manual, 'Main Power Wiring – 3-Phase Power Systems' on page 3-2.

Electrical Connections

Single-Phase, Main Power Connection



Rinse Signal Wiring



Detergent Signal Wiring

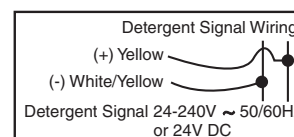


Figure QS-4 Wiring Diagrams

Description of Controls

The keypad shown in Figure QS-5 displays and controls dispenser functions. Blinking characters prompt you to input a value with the **SCROLL** key. Use the **NEXT** key to move to the next digit. When the correct value displays, press the **ENTER** key to save and return to the main menu.



Figure QS-5
Controller Keys

Next Scroll Enter

Key	Description
NEXT	Displays User or Program menus. Moves through each screen that does not have a blinking digit.
SCROLL	Moves between setting choices within a menu. Changes the blinking number displayed (0-9). Starts Probeless Manual Initial Detergent Charge.
ENTER	Views and changes settings. Sets value and exits program. Starts and stops pumps and prime.

User Mode Operation

In User Mode, the Sprite III (Alpha) displays operating status, alarms, and gives access to the Program Mode menus:

User Mode Menus & Functions

- Rack Count (*r R C*) in multiples of 20 racks
- Detergent Prime (*d P*)
- Rinse Pump Prime (*r P*)
- Third Product Pump Prime (*3 P*)
- De-lime (*d L*)
- ID Mode
- Live Conductivity Display (if enabled)
- Password Access to Program Mode
- Audible & Visible Alarms (Low Detergent, Detergent Overfeed Stop, Change Tank Water)

Alarms and Indicators

These indicators and alarms may be seen in User Mode.

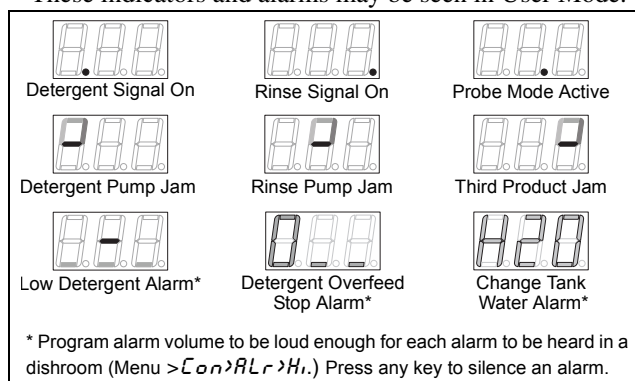


Figure QS-6 Alarms and Status Indicators

Low Detergent Alarm (Probe Mode Only)

This audible and visual alarm occurs when all of the following conditions are met:

1. A detergent signal occurs, AND
2. The conductivity level falls below the programmed set point, AND
3. The conductivity level remains low for the number of racks set in "Low Detergent Alarm Delay" (*LdA*).

Reset Low Detergent Alarm

Conditions that cause the alarm to reset are:

1. Conductivity rises by 10% (OR rises to the set point) within the number of racks set in *LdA*, OR
2. A key is pressed on the controller (pause), OR
3. Device power is cycled off and on (alarm will resume unless conductivity rises).

Detergent Overfeed Stop Alarm (Probe Mode Only)

This audible and visual alarm occurs if a low detergent alarm condition continues for twice the preset number of racks washed (as programmed in the Low Detergent Alarm Delay menu *>AL.P>LdA*). At that point, the detergent feed stops. To begin detergent feed after changing the product container, press any key.

Change Tank Wash Water Alarm

This audible and visual alarm alerts the operator to drain the wash tank and refill it with fresh water. It occurs after a set number of racks run. This alarm must be enabled in "Change Tank Wash Water Alarm" (Menu *CnF>CLtA*).

Rack Count

Racks are counted by 20s. Each time the count reaches *999*, the display automatically rolls over to *000* and a decimal lights up. The display can rollover a maximum of three times; each rollover causes another decimal to light.

One lit decimal indicates approximately 20,000 racks have run. Two lit decimals indicate approximately 40,000 racks have run. Three lit decimals indicate approximately 60,000 racks have run. When *999* plus three lit decimals

display, 80,000 racks have run and no additional racks can be counted until the Rack Counter is reset in Menu *CnF>rCr*.

User Mode Menus

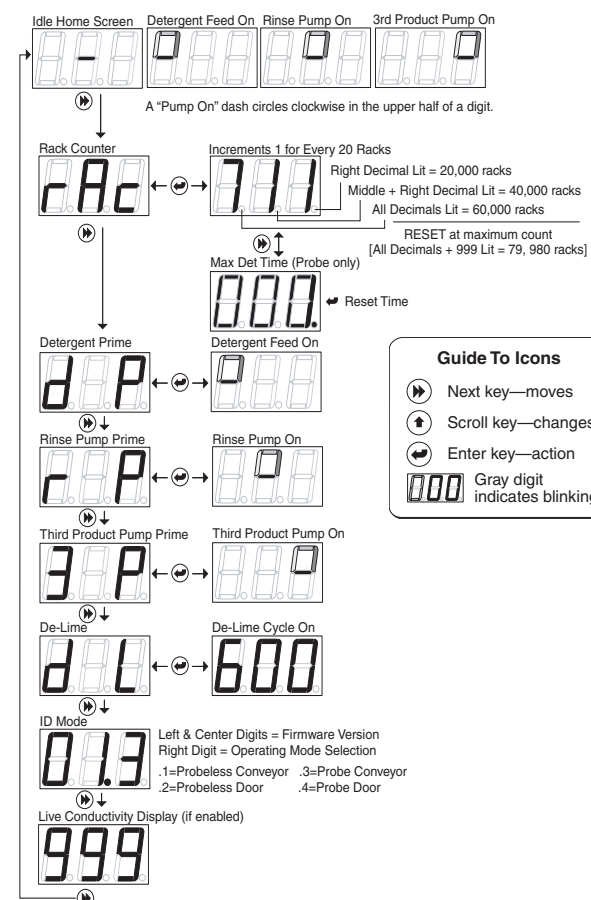


Figure QS-7 User Mode Menus & Pump Functions

Password Access to Program Mode

A password is needed to enter Program Mode. The factory set password (*123*) can later be changed to any 3-digit password.

Note

Make note of a new password. If your password is lost, you cannot program the unit and must contact Customer Service for assistance.

1. Press and hold **ENTER** for two seconds to access the password input screen. Three zeros display.
2. Press **SCROLL** to change the blinking digit to a 1, then press **NEXT** to move to the next digit.
3. **SCROLL** until 2 displays in the center digit.
4. Continue until *123* displays, then press **ENTER** to reach Program Mode's Configuration Menu (*CnF*).

Program Mode Notes

Menus are illustrated in Figure QS-8 on page 5. Some menus will not appear, based upon Configuration Settings. Configuration Menus (*CnF*) and one Adjustment Menu, Probe (*AL.P*) or Probeless (*AL.n*), display.

NOTE: In new installations, set Configuration Settings first.

Press **ENTER** to view the first selection, then **SCROLL** to view additional selections or to set the value. Press **ENTER** to “save” or press **NEXT** to exit without saving.

Configuration Settings (Menu *CNF*)

Detergent Control *dcr*: Probe or No Probe

Machine Type *mt*: Door or Conveyor

Detergent Type *dt*: Liquid or Capsule

Pulse Feed *Pul* (Capsule Detergent & Probe Mode Only)

Left digit = Seconds On; Center digit = Seconds Off;

Right digit = % of setpoint (5 = 50%, 8 = 80%, etc.)

Alarm Volume *ALr*: Off, Low, Medium or High

Rack Time in Rinse *rtr* (Conveyor-type machine only): Set the number of seconds it takes to move one rack a single rack length’s distance.)

Rack Count Reset *rCr*: Press **ENTER** to display the count, **SCROLL** to display *CLr*, then **ENTER** to reset count to 0.

Automatic Initial Charge *Aic* (Probeless Mode only):

❖ Off = User starts Manual Charge with **SCROLL** key.

❖ On, Door machine = Automatic charge occurs each time dispenser receives a rinse signal that was not preceded by a detergent signal within 90 seconds. A five minute lockout timer begins at start of charge.

❖ On, Conveyor machine = Automatic charge occurs each time the dispenser detects a detergent signal input that remains on for 10 seconds continuously.

Change Installer Password: Change from the default password to any 3-digit password; note new password.

Change Tank Wash Water Alarm *CLr*: Number of individual racks to run (1-999, 0=Off) before alarm activates.

3rd Product Pump: Run with Rinse or Detergent pump.

Adjustments – Probe Mode (Menu *Adj.P*)

Detergent Tank Reading *dtr*: Press **SCROLL** to add product to proper concentration, then test. Solution must be well mixed (wash pump running) and at operating temperature. Press **ENTER** to view reading (Detergent Setpoint).

Detergent Setpoint *dSP*: Setpoint is most accurate when used with new Nova Controls probe (provided).

Live Conductivity Display *Lcd*: On or Off

Rinse Pump Speed *rPS*: See RPM Settings Guide, below. With standard rinse tube, dispenses .5 ml/rev.

3rd Product Pump Speed *3PS*: At maximum speed, dispenses 6 oz. / min. Set to 0 to turn pump off.

Low Detergent Alarm Delay *LdA*: Number of racks to run before Low Detergent Alarm is activated.

Rinse Feed Option *rFo*: Determines whether rinse pump is activated by the rinse signal or by the detergent signal.

Rinse Delay Time *rdt* (Door, rinse on rinse only): Restricts rinse injection to last seconds (0-19) of each rack.

Adjustments – Probeless Mode (Menu *Adj.n*)

Detergent Initial Charge *dic*: Time needed to charge wash tank to correct concentration on an initial fill.

Detergent Dose *dos*: Detergent pump run time (0-59 secs.)

Detergent Dose Interval *ddi*: Dose every rack (1), every other rack (2), or every third rack (3).

Rinse Pump Speed *rPS* & 3rd Product Pump Speed *3PS*: See instructions in Adjustments – Probe Mode.

Rinse Feed Option *rFo*: Determines whether rinse pump is activated by the rinse signal or by the detergent signal.

Rinse Delay Time *rdt* (Door, rinse on rinse only): Restricts rinse injection to last seconds (0-19) of each rack.

RPM Settings Guide for Rinse / 3rd Product Pumps

(Use Guide as a starting point, then adjust rinse additive, as necessary.)

Final Rinse Time (Sec.)	1 ml/rack	1.5 mls/rack	2 mls/rack	2.5 mls/rack	3 mls/rack	3.5 mls/rack	4 mls/rack	4.5 mls/rack	5 mls/rack
6	20	30	40	50	60	--	--	--	--
8	15	23	30	38	45	53	60	--	--
10	12	18	24	30	36	42	48	54	60
12	10	15	20	25	30	35	40	45	50
14	9	13	17	21	26	30	34	39	43
15	8	12	16	20	24	28	32	36	40
16	8	11	15	19	23	26	30	34	38
18	7	10	13	17	20	23	27	30	33
20	6	9	12	15	18	21	24	27	30

Limited Warranty

SELLER warrants solely to BUYER the Products will be free from defects in material and workmanship under normal use and service for a period of one year from the date of completion of manufacture. This limited warranty does not apply to (a) hoses; (b) and products that have a normal life shorter than one year; or (c) failure in performance or damage caused by chemicals, abrasive materials, corrosion, lightning, improper voltage supply, physical abuse, mis-handling or misapplication. In the event the Products are altered or repaired by BUYER without SELLER’S prior written approval, all warranties will be void. NO OTHER WARRANTY, ORAL, EXPRESS OR IMPLIED, INCLUDING ANY WARRANTY OF MERCHANTABILITY OR FITNESS FOR ANY PARTICULAR PURPOSE, IS MADE FOR THESE PRODUCTS, AND ALL OTHER WARRANTIES ARE EXPRESSLY EXCLUDED.

SELLER’S sole obligation under this warranty will be, at SELLER’S option, to repair or replace F.O.B. SELLER’S facility in Cincinnati, Ohio any Products found to be other than as warranted.

Limitation of Liability

SELLER’S WARRANTY OBLIGATIONS AND BUYERS REMEDIES ARE SOLELY AND EXCLUSIVELY AS STATED HEREIN. SELLER SHALL HAVE NO OTHER LIABILITY, DIRECT OR INDIRECT, OF ANY KIND, INCLUDING LIABILITY FOR SPECIAL, INCIDENTAL, OR CONSEQUENTIAL DAMAGES OR FOR ANY OTHER CLAIMS FOR DAMAGE OR LOSS RESULTING FROM ANY CAUSE WHATSOEVER, WHETHER BASED ON NEGLIGENCE, STRICT LIABILITY, BREACH OF CONTRACT OR BREACH OF WARRANTY.

Program Mode Menus

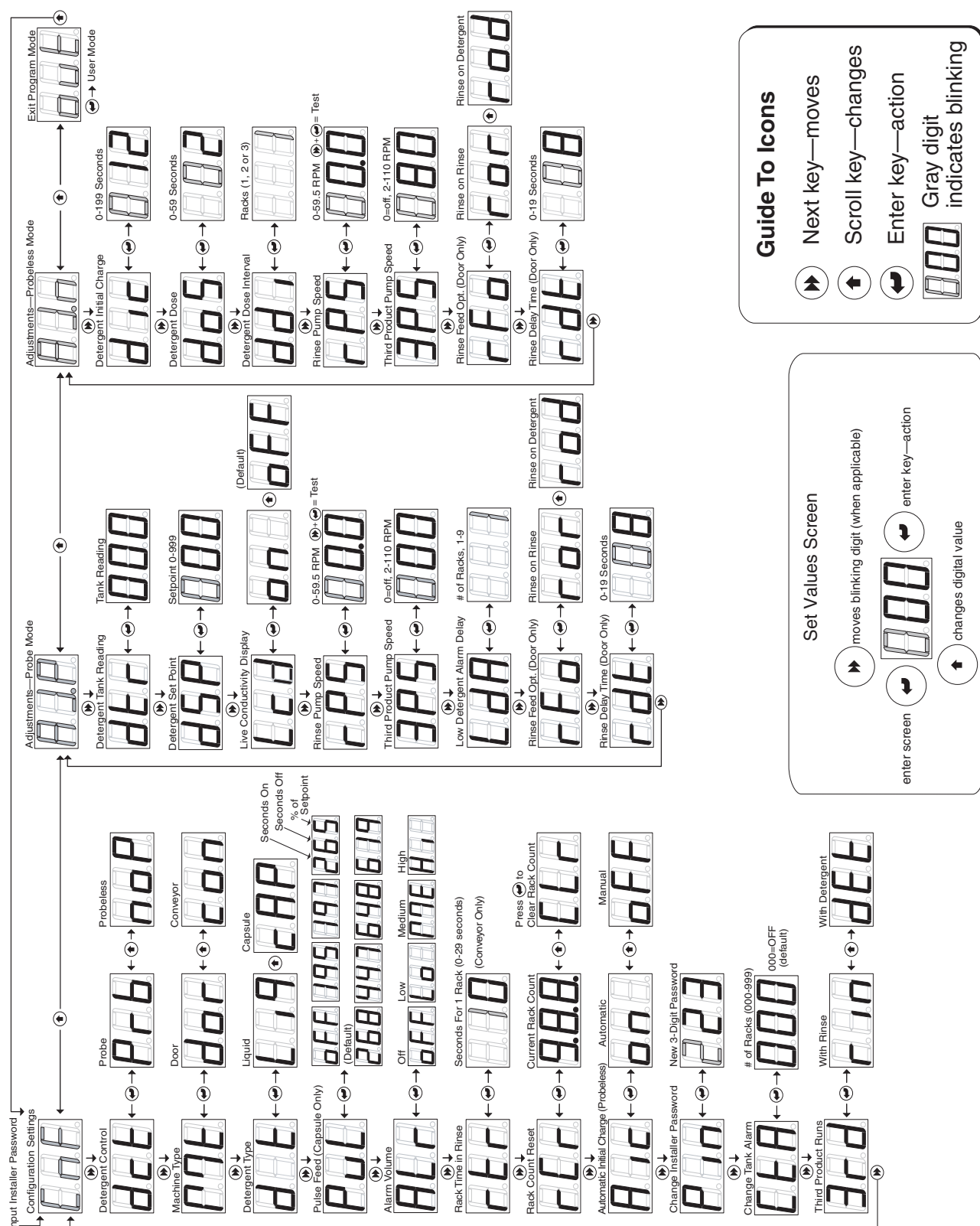
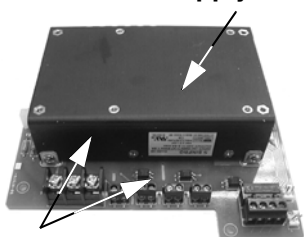


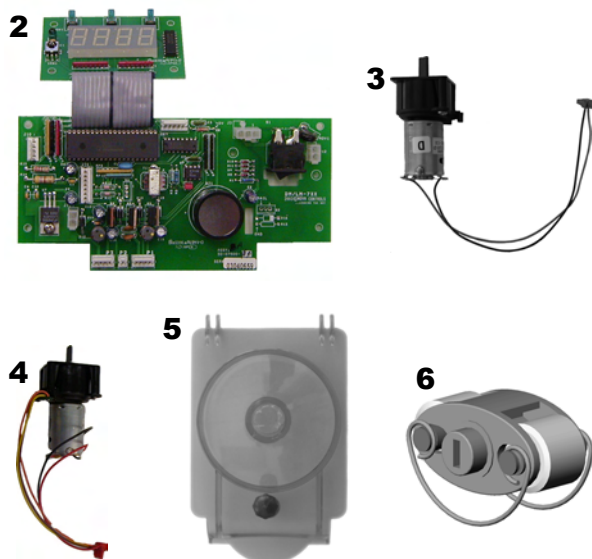
Figure QS-8 Program Mode Menus

Spare Parts Listing

1a. Power Supply Module Only



**1b. Power Supply Assembly
(includes module, PCB, harness)**



Specifications

Size	8" W x 9" H x 6" D (20.3 cm W x 23 cm H x 15.4 cm D)
Weight	6 lbs. (2.72 kg)
Temperature	10° to 40° C (50° to 120° F) Maximum. See the Sprite III Alpha Reference manual for a complete list of Environmental Specifications.
Power	Total amperage draw during operation is: 100 to 240 VAC nominal (+/- 10% fluctuation), 50/60 Hz., 1.0 amps (max)
Rinse Pump Flow Rate	0.5 mls/revolution
Detergent Pump Flow Rate	6 oz./min. (178 mls/min)
3rd Product Pump Flow Rate	0.60 oz./min. to 6 oz./min. (17.8 mls/min. to 178 mls/min.)

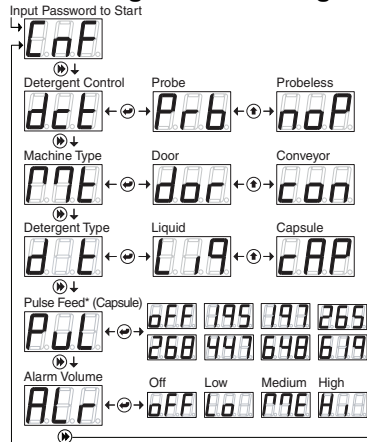
Ref. #	Part Number	Description
1a	13-07724-00	Power Supply Module Only
1b	13-07724-01	Power Supply Assembly (includes module, PCB and Harness)
2	13-07725-10	Logic Circuit Board
*	13-07725-01	Logic Circuit Board w/ Ext. Alarm Option
3	13-06524-01	Pump Motor Detergent w/2-Pin Plug
4	13-06523-01	Pump Motor Rinse w/5-Pin Plug
5	13-06398-00	Pump Front, RED, w/Captive Screw Add'l. Colors (-04 BLUE) (-05 ORANGE) (-11 SMOKE)
6	13-06396-00	Pump Spinner Assembly (all pumps)
*	37-07584-02	Cabinet Front, LT. GRAY (-05 BLACK)
*	37-07585-32	Cabinet Rear, LT. GRAY (-05 BLACK)
*	13-07268-06	Power Switch Cover
*	13-07122-00	Dispenser Mounting Bracket Kit
*	13-07191-00	Solenoid Valve Kit, 1/4" Fitting
*	13-07191-02	Solenoid Valve Kit, 5/16" Fitting
*	13-06528-00	Solenoid Valve Nut w/Sleeve, 1/4" 10 pk.
*	13-06528-02	Solenoid Valve Nut w/Sleeve, 5/16" (8mm) 10 pk.
*	13-06395-10	Pump Tube, Rinse/Sanitizer Lo-Flow, EPDM (incl. 1/4" Fittings w/o nuts) 10 pk.
*	13-06395-20	Pump Tube, Rinse / Sanitizer Lo-Flow, 10 pk. EPDM (incl. 1/4" and 1/8" Fittings w/o nuts)
*	13-06395-30	Pump Tube, Rinse / Sanitizer Lo-Flow, EPDM (incl. 1/8" Fittings w/o nuts) 10 pk.
*	13-06928-10	Pump Tube, Rinse / Sanitizer Lo-Flow, Silicone (incl. 1/4" Fittings w/o nuts) 10 pk.
*	13-06928-20	Pump Tube, Rinse / Sanitizer Lo-Flow, 10 pk. Silicone (incl. 1/4" and 1/8" Fittings w/o nuts)
*	13-06928-30	Pump Tube, Rinse / Sanitizer Lo-Flow, Silicone (incl. 1/8" Fittings w/o nuts) 10 pk.
*	13-06399-10	Pump Tube, Detergent / Sanitizer, EPDM (incl. 1/4" Fittings w/o nuts) 10 pk.
*	13-06563-00	Pump Tube Compression Nut w/Sleeve, 1/4", 10 pk.
*	13-06563-02	Pump Tube Compression Nut and Ferrule, 1/8" (3mm) 10 pk.
*	13-06529-00	Rinse Injection Fitting, Straight, Kynar, 1/8" NPT x 1/4" Tube
*	13-06529-02	Rinse Injection Fitting, Straight, Kynar, 1/8" NPT x 6mm Tube
*	13-06531-00	Rinse Injection Fitting, 90°, Elbow, 1/8" NPT x 1/4" Tube
*	13-06531-01	Rinse Injection Fitting, 90°, Elbow, 1/8" NPT x 1/8" (3mm) Tube



Quick Reference Wallet Card

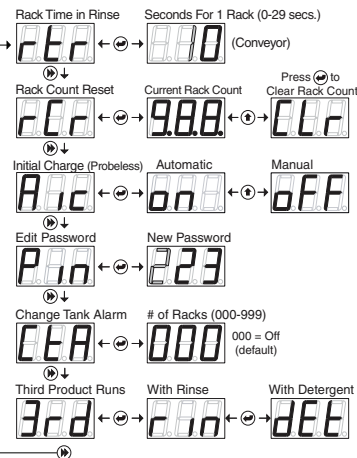
Cut along dotted outer line and keep for future reference.

Configuration Settings



* NOTE: Pulse Feed display is ONLY available when Detergent Type = Capsule (Menu d t = cRP)
195 = 1 Second On, 9 Seconds Off, 50% of Set Point
197 = 1 Second On, 9 Seconds Off, 70% of Set Point
265 = 2 Seconds On, 6 Seconds Off, 50% of Set Point
268 = 2 Seconds On, 6 Seconds Off, 80% of Set Point (Default)
447 = 4 Seconds On, 4 Seconds Off, 70% of Set Point
648 = 6 Seconds On, 4 Seconds Off, 80% of Set Point
619 = 6 Seconds On, 1 Second Off, 90% of Set Point

Configuration Settings con't.



Sprite III (Alpha) Warewash Quick Reference Card

20-09092-00 Rev. C

Hydro Systems Company

www.hydrosystemsco.com

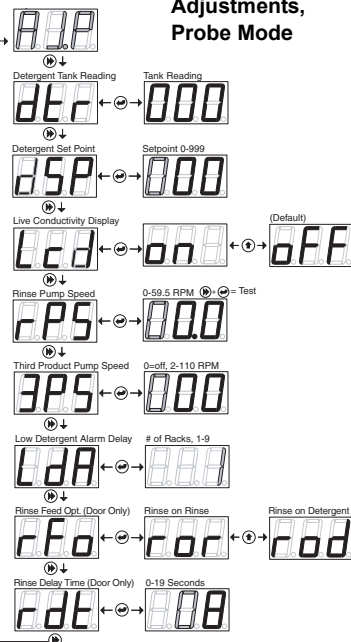
A DOVER COMPANY

3798 Round Bottom Road, Cincinnati, OH 45244 USA • Phone (513) 271-8800 • Fax: (513) 271-0160

Fold Here

Fold Here

Adjustments, Probe Mode



Adjustments, Probeless Mode

